

Modern decisions are often limited by how efficiently we can extract useful structure from data. Sometimes data are limited or incomplete; sometimes the dataset or required computation is too large to process. My research targets one question across both settings: **what information or computation is necessary for a reliable decision?** My two research threads develop practical, provable strategies for recovering what is missing and compressing what is redundant—without changing the resulting inference or action.

Decision-making with incomplete information Many decision-making problems involve panel data: entries encode outcomes or measurements across units, time points, and interventions, alongside covariates, missingness, and confounding. Yet the quantity needed for a decision—often a unit’s outcome under an unchosen intervention—is unobserved. The challenge is to recover it with valid uncertainty using enough structure to identify the target.

My group develops practical methods with guarantees for granular, personalized inference. We began with nearest-neighbor (NN) unit-by-time counterfactual inference under nonparametric factor models.^[14] We sharpened it through doubly robust NN inference^[13] and cross-fitted inference combining matrix completion, denoising, and inverse weighting.^[1] We then established that two-sided NN adapts to unknown smoothness and achieves minimax rates.^[22] Temporal latent dynamics enabled counterfactual forecasting,^[9] while GAAVI developed nonparametric anytime-valid inference.^[7] A complementary line extended scalar inference to distributions through kernel embeddings,^[8] Wasserstein barycenters,^[15] and exponential-family models under unobserved confounding.^[23]

Looking ahead, my vision is a default *tuning-free* method that discovers panel structure, adapts automatically, and returns valid uncertainty. We are pursuing two routes. One develops adaptive NN procedures that eliminate manual tuning and cross-validation.^[21] The other follows the *bitter lesson*^[25]: it uses computation-heavy pretraining rather than encoding increasingly elaborate priors in modeling assumptions. TabImpute provides the starting point, using synthetic data to train a transformer for zero-shot imputation across heterogeneous tables and missingness patterns.^[16] We plan to bridge these approaches into a unified recipe that selects between—or combines—a lightweight structure-adaptive estimator and a computation-heavy learned model according to the problem at hand.

Structure-aware computation for faster inference Many inferential methods are statistically accurate but computationally prohibitive at modern scale. Naive shortcuts reduce cost by discarding data or approximating computation, but can distort estimates and invalidate uncertainty guarantees. My group develops structure-aware compression methods that reduce computation while preserving the fidelity of the final inferential answer.

Kernel thinning (KT) began this program by compressing a distribution while preserving integration accuracy,^[12] and generalized KT extended the guarantees across kernels and distributions.^[11] Compress++ made high-quality distribution compression near-linear in the input size,^[24] while Stein KT handled biased inputs such as non-converged Markov chains.^[20] Together, they support a *Compress-then-X* framework with near-quadratic gains and almost no loss of inferential accuracy in hypothesis testing,^[10] kernel-based learning,^[19] neural-network training,^[4] interpretability,^[3] experimental design,^[2] and mean-field dynamics.^[5]

Next, I want to build a theory-guided compression strategy for the entire large language-model (LLM) pipeline. As contexts and agent calls grow, memory and computation become binding constraints. Existing efforts focus on hardware, numerical precision, and token management; our complementary approach is statistical: retain only the information needed for the final prediction or action. Thinformer takes a first step by introducing a noncausal compression scheme for approximating attention.^[4] Express extends this idea to causal compression, accelerating long-context prefill, KV-cache compression, and memory- and compute-constrained long-form decoding.^[17] My goal is to build a practical recipe spanning inference, training, and agent execution—compressing examples, gradients, and intermediate states during training, and growing histories and tool interactions during agent execution.

From theory to measurable, reproducible progress The last two decades of progress in statistics, machine learning, and AI have shown that progress accelerates when theory and algorithms are coupled with shared benchmarks and open implementations. We are building this infrastructure alongside our methods: N²-Bench for nearest-neighbor completion,^[6] MissBench for tabular imputation,^[16] and ORCA-bench for agentic root-cause analysis,^[18] alongside open-source N², GoodPoints, TabImpute, Thinformer, and Express. By pairing theory with benchmarks and software, we hope to make advances in inference and AI faster and more reliable, while keeping progress measurable, reproducible, and open.

References

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